

arsenide (GaAs), as it has a band gap of 1.43eV, which is extremely close to the optimal band gap range. It has six times higher electron mobility than silicon, which allows faster operation. It is used in conjunction with indium gallium phosphide (InGaP) and ger-

manium (Ge) layer, with bandgaps of 1.85eV and 0.67eV, respectively.

InGaP utilises high energy photons, while Ge utilises much lower energy photons, and GaAs utilises those in between. Increasing the number of junctions increases the efficiency of the cell. But the numbers can't be increased as per will. There is an upper limit.

The increase in film thickness decreases the transmittance of light, and photons may not make it to the bottom layers of the cell. The inverse also has a limit. If the semiconductor films are too thin, these will transmit too much light and will not absorb all the photons.

In MJ solar cell, lattice structures of adjacent layers have a bearing on its efficiency. Layers can either be lattice-matched or lattice-mismatched, as shown in Figs. 5 (a) and 5 (b), respectively.

The mismatch of the lattices produces a lower band gap as opposed to that of the matched lattices. With the addition of an active Ge subcell, upright three-junction solar cells with lattice-matched InGaP/InGaAs/Ge designs have achieved more than thirty per cent efficiency under one sun illumination and 41.6 per cent efficiency under 364 suns.

The Ge subcell makes incomplete use of higher energy photons as carriers. This problem can be

addressed by dividing the solar spectrum more finely using cells with five, six, or even more junctions. The subcell bandgaps needed for these MJ cell designs can be accessed easily with III-V materials that are not all at the same lattice constant.

However, such lattice-mismatched or metamorphic materials typically have crystal dislocations that act as recombination centres. Special techniques during manufacturing are used to reduce crystal imperfections.

For terrestrial applications, the high cost of these semiconductor substrates may be offset by using concentrating optics, primarily using Fresnel lenses. Fresnel lenses are shaped like a dartboard, with concentric rings of prisms around a lens that is a magnifying glass and placed on top of a solar cell, as shown in Fig. 6.

These focus scattered light to a tight beam. The concentrating optics increases the amount of light incident on the solar cell, thus leading to more power production. Concentrating optics allows individual cells to be quite small—at times as small as the size of a pencil tip. Therefore this technique allows hundreds of solar cells to be grown in a single batch. This decreases the number of solar cells needed and hence cost.

Architecture of multi-junction solar cell

The architecture of an MJ solar cell and a graph showing its utilisation of the solar spectrum is shown in Fig. 7.

The function of an emitter window layer is to render passivity in the emitter surface. In any solar cell, a high surface recombination velocity will reduce the photoresponse of the solar cell, mostly in the blue portion of the solar radiation spectrum. AlInP is generally used as a window layer. It provides reasonable confinement in the emitter of an n-on-p device. It upgrades the blue response and fill factor of the solar cell.

Tunnel junctions act like diodes in reverse polarity to the original configuration in the stack of subcells. These provide better optical, electrical, and mechanical matching between the subcell layers.

The tunnel junction between subcell layers provides a low resist-

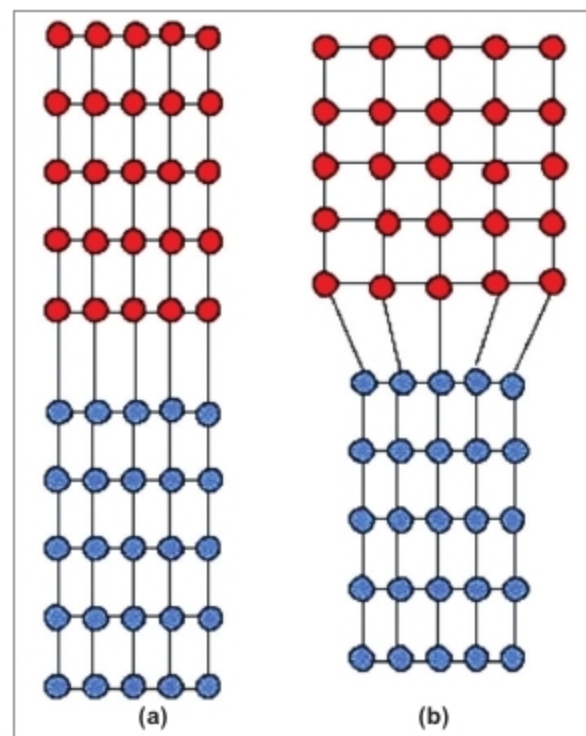


Fig. 5: Lattice structure (a) Lattice matched, (b) Lattice mismatched

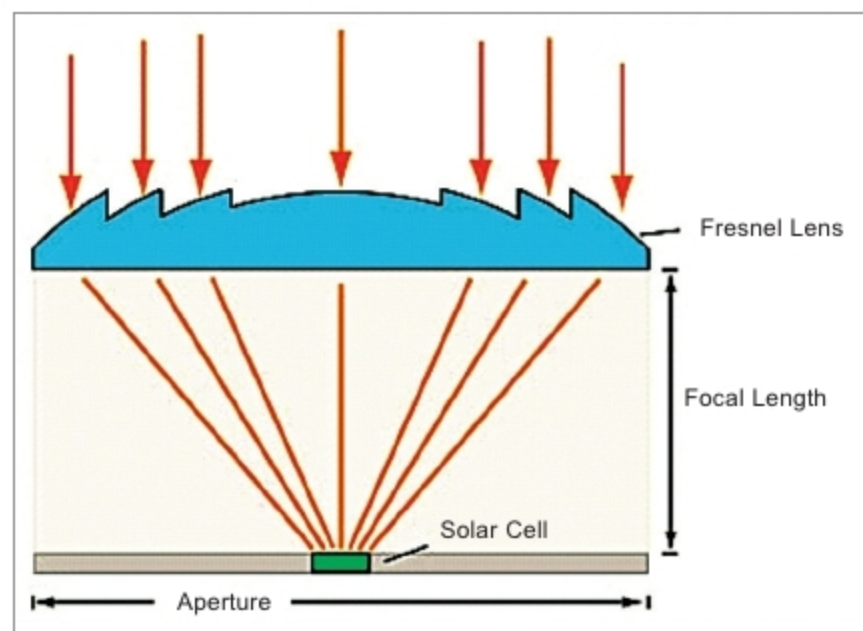


Fig. 6: Schematic diagram of Fresnel lens

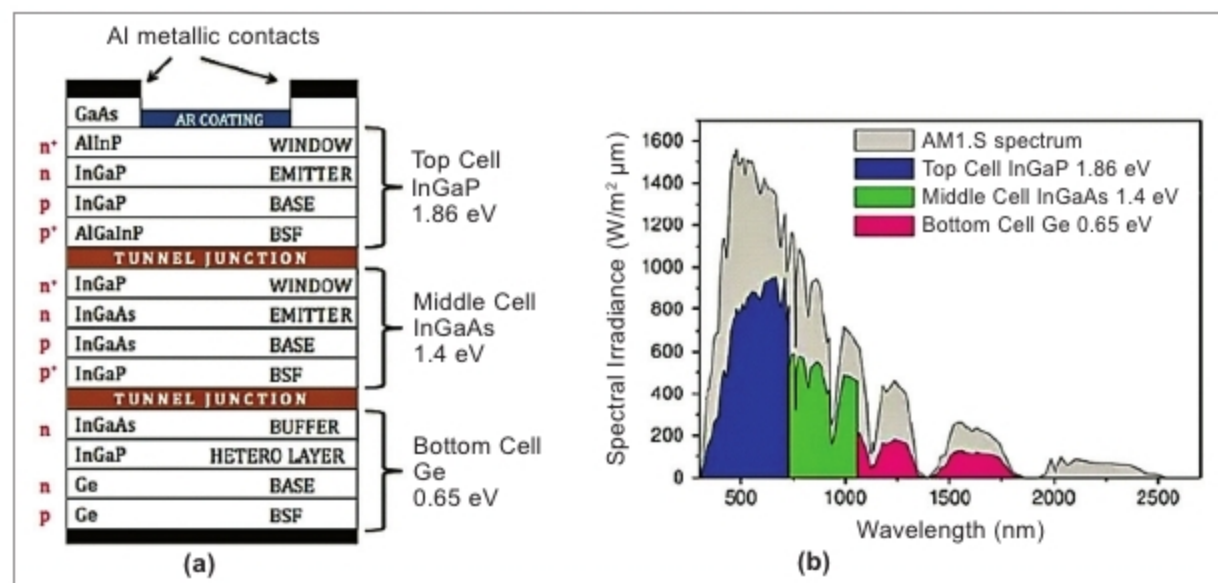


Fig. 7: Architecture of a multi-junction solar cell and its solar spectrum utilisation